OSV Construct
System Build Spec

Full specification for developer — Lead generation, quoting, AI pricing, ticketing & job management

MELBOURNE, VIC | CONSTRUCTION & TRADES | FULL STACK | AI-POWERED | MOBILE-FIRST

00 — OVERVIEW What We're Building

A unified operations platform for **OSV Construct** — a Melbourne-based residential construction and trades business. The system replaces disconnected tools (WhatsApp quotes, paper jobs, spreadsheets) with a single AI-powered platform covering the full lifecycle: **lead in → quote → job → invoice → done.**

PHASE 1 — CORE

Lead + Quote + Ticket

Capture leads, generate AI quotes, create customer tickets, send to client for approval

PHASE 2 — OPERATIONS

Job + Crew + Comms

Assign crew, schedule jobs, track progress, client portal, change orders

PHASE 3 — INTELLIGENCE

Analytics + AI Learning

Quote win/loss analysis, AI pricing refinement, lead source ROI, pipeline forecasting

ℹ️ **Tech Stack Recommendation**

Frontend: React + Tailwind. Backend: Node.js / Express or Laravel. Database: PostgreSQL. AI: OpenAI API (GPT-4o) for quote generation. Auth: Clerk or Supabase Auth. File storage: Cloudflare R2 or AWS S3. SMS/Email: Twilio + Resend.

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01 — USERS User Roles & Permissions

👷

OWNER / ADMIN

Full access. Jack (OSV Construct). Manages leads, approves quotes, views all jobs, controls pricing, sees all financials and analytics.

🔨

TRADESPERSON

Field access via mobile. Views assigned jobs. Creates quotes on site. Attaches photos and notes. Updates job status. No financial data.

🏠

CLIENT

Read-only client portal. Receives and approves quotes. Tracks job status. Views invoices. Submits variation requests. No internal data.

PERMISSION MATRIX

ACTION	OWNER	TRADIE	CLIENT
View all leads	✅	❌	❌
Create a quote	✅	✅ (own jobs)	❌
Approve quote before sending	✅	❌	❌
Accept/reject a quote	✅	❌	✅
View job details	✅	✅ (assigned)	✅ (own jobs)
Update job status	✅	✅	❌
View pricing/margins	✅	❌	❌
Edit price book	✅	❌	❌
Send invoices	✅	❌	❌
Request variations	✅	✅	✅

02 — ARCHITECTURE

System Architecture

```
// Core data entities and their relationships

Lead          → can become →   Ticket
Ticket        → generates →    Quote
Quote         → approved →     Job
Job           → completed →    Invoice
Invoice       → paid →         Payment

// Supporting entities
PriceBook     → feeds →        Quote line items
Client        → has many →     Tickets, Quotes, Jobs
User (Tradie) → assigned →     Jobs
Variation     → attached →     Job (mid-work changes)
```

03 — LEADS

Lead Capture System

Every lead that enters the business must be captured into the system regardless of source. No lead should be managed via text or memory.

LEAD SOURCES — ALL MUST FEED INTO SYSTEM

Website Contact Form

Embeddable form on OSV Construct website. Auto-creates a lead record on submit. Required fields: name, phone, email, suburb, job type, description, photos (optional).

📱 **Manual Entry (Owner/Tradie)**

Quick-add lead form in the app. Used for phone calls, word of mouth, door knocking, or referrals. Minimum: name + phone + job type.

📧 **Email Parse (Optional Phase 2)**

Parse inbound emails from a designated address (leads@osvco.au) and auto-create lead records from the content.

📣 **Facebook / Instagram Leads**

Manual import or webhook from Meta Lead Ads. Map form fields to lead schema on creation.

LEAD RECORD — REQUIRED FIELDS

FIELD	TYPE	NOTES
lead_id	UUID	Auto-generated
client_name	String	First + last
phone	String	Primary contact
email	String	Optional but needed for quote delivery
suburb	String	For job site location
job_type	Enum	Decking, Fencing, Framing, Landscaping, Carpentry, Pergola, Other
description	Text	Client's description of the work needed
photos	File[]	Optional site photos uploaded at enquiry

source	Enum	Website, Referral, Facebook, Instagram, Google, Word of mouth, Other
status	Enum	New → Contacted → Site visit → Quoted → Won → Lost
created_at	Timestamp	Auto
assigned_to	User ID	Owner or tradie handling the lead

⚠️ **Speed to Contact is Critical**

Build an automated SMS/notification that fires the moment a new lead is created. Studies show response within 5 minutes increases conversion by over 400%. The system must alert the owner immediately via push notification and optionally auto-send a holding SMS to the client.

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04 — PIPELINE Lead Pipeline & Status Tracking

PIPELINE STAGES

- New — Lead just received. Not yet contacted. Triggers instant alert to owner.
- Contacted — First contact made. Date/time logged. Notes added.
- Site Visit Scheduled — Site inspection booked. Date/time stored. Linked to calendar.
- Site Visit Done — Visit completed. Photos and measurements attached. Quote pending.
- Quoted — Quote sent to client. Quote ID linked. Awaiting response.
- Follow-Up Required — No response after X days. System flags and prompts

follow-up.

- Won — Quote accepted. Auto-creates ticket/job. Lead archived.
- Lost — Lead declined or gone cold. Reason logged. Used for analytics.

PIPELINE VIEW (OWNER DASHBOARD)

Kanban Board — Required

- Visual kanban board with one column per stage
- Lead cards show: client name, suburb, job type, days in current stage, quote value if quoted
- Drag and drop between stages (updates status in DB)
- Filter by: job type, suburb, date range, assigned user
- Total pipeline value shown at top (sum of all quoted leads)
- Red highlight on cards sitting in same stage for 5+ days with no activity

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05 — QUALIFY Lead Qualification

QUALIFICATION CHECKLIST — BUILT INTO LEAD FORM

These fields must be completed before a lead can be moved to "Quoted" status:

- Budget range confirmed? — Under 2k / 2–5k / 5–15k / 15k+ / Unknown
- Timeline — ASAP / 1 month / 3 months / Planning only

- Decision maker? — Yes / No / Shared (owner + partner)
- Site accessible? — Yes / Need to confirm / No access issues noted
- Council approval needed? — Yes / No / Unknown
- Competing quotes? — Yes / No / Unknown

> 🚩 **Auto Red-Flag Conditions**
> If budget is "Unknown" + timeline is "Planning only" + decision maker is "No" — flag the lead as low priority automatically. Owner sees a red badge on the lead card. Don't spend quoting time until qualified.

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06 — NURTURE

Follow-Up & Nurture System

AUTOMATED FOLLOW-UP SEQUENCE

TRIGGER	DELAY	ACTION	CHANNEL
New lead created	Instant	Push notification to owner + optional holding SMS to client	App + SMS
Quote sent, no response	48 hrs	Reminder to owner to follow up. Draft follow-up SMS generated by AI.	App alert
Quote sent, no response	5 days	Second follow-up prompt. System drafts a "checking in" message.	App alert
Quote sent, no response	14 days	Mark lead as "Gone Cold". Prompt owner to close or extend.	App alert
Job	7 days	Auto-send review request SMS/email to	SMS +

completed	after	client.	Email

✅ **AI Follow-Up Message Drafting**

When a follow-up is triggered, the system should pre-generate a draft SMS/email using the client's name, job type, and suburb. Owner reviews and sends with one tap. Not fully automated — owner always approves before send.

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07 — QUOTING

Quote Creation System

Quotes can be created from a lead, from a ticket, or as a standalone. The creation process must work on desktop and mobile in the field.

QUOTE CREATION FLOW

📋 Select Lead / Client → 🏷️ Job Details → 🤖 AI Generate → ✏️ Review & Edit →

QUOTE LINE ITEM STRUCTURE

```
// Each quote contains one or more line items
{
  quote_id:        "UUID",
  lead_id:         "UUID (optional link)",
  ticket_id:       "UUID (optional link)",
  client_id:       "UUID",
  quote_number:    "OSV-2026-0042",
  status:          "Draft | Sent | Approved | Declined | Expired",
  valid_until:     "Date (default 30 days)",
  line_items: [
```

```
    {
      item_id:      "UUID",
      category:     "Labour | Materials | Disposal | Equipment",
      name:         "Merbau Decking Supply & Install",
      description: "Supply and install 90x19mm Merbau decking boards...
      qty:          24.5,
      unit:         "m2 | lm | item | hrs",
      unit_price:   185.00,
      total:        4532.50,
      taxable:      true
    }
  ],
  subtotal:      number,
  gst:           number (10%),
  total_inc_gst: number,
  notes:         "String — terms, exclusions, site access requirements"
  internal_notes:"String — NOT visible to client",
  margin_pct:    number — owner view only
}
```

PRICE BOOK — REQUIRED

Master Price Book

A database of pre-set line items the owner manages. Used by AI to generate quotes and by tradies to select items quickly in the field.

- **Categories:** Labour, Materials, Disposal, Equipment, Subcontractor
- **Each item has:** name, description template, unit, default unit price, cost price (hidden), default margin %
- **Owner can:** add, edit, delete, and set prices. Prices auto-update in future quotes but never retroactively change sent quotes
- **AI uses:** price book as the base for quote generation. Falls back to Melbourne market rates if item not in price book
- **Import/export:** CSV import to bulk load items

08 — AI ENGINE

AI Pricing & Quote Generation

🤖 **How AI Quote Generation Works**

Owner or tradie enters: job type, dimensions/scope, site notes, photos (optional). System sends this to GPT-4o with the price book, Melbourne labour rates context, and a structured prompt. AI returns a full list of line items with quantities, descriptions, and prices. Owner reviews and edits before sending.

AI SYSTEM PROMPT STRUCTURE

```
// System prompt sent to GPT-4o for quote generation

You are a construction estimator for OSV Construct, a Melbourne-based
trades business. Generate a detailed, accurate quote based on:

Job Type:     {job_type}
Description:  {client_description}
Dimensions:  {measurements_if_provided}
Site Notes:   {tradie_notes}
Price Book:   {relevant_price_book_items_JSON}

Rules:
- Use Melbourne 2026 labour and material rates
- Separate Labour, Materials, Disposal line items
- Include site prep and access considerations
- Flag if council approval may be required
- Return ONLY valid JSON — no markdown, no explanation

Return format:
{
```

```
  "line_items": [...],
  "scope_summary": "2-3 sentence plain English summary for client",
  "exclusions": ["list of what is NOT included"],
  "assumptions": ["list of assumptions made"],
  "estimated_days": number,
  "flags": ["any risks or items needing owner review"]
}
```

AI FEATURES REQUIRED

📝 **Auto Line Item Generation**

AI breaks a job description into individual billable line items with quantities, units, and prices pulled from price book or market rates.

📄 **Scope Summary Writing**

AI generates a professional plain-English scope summary for the client-facing quote. 2–4 sentences. Clear, confident, trades-authentic tone.

⚠️ **Risk & Flag Detection**

AI flags items that may need owner attention — council permits, structural concerns, access issues, scope ambiguity — before quote is sent.

🔄 **Variation Drafting**

When a change order is raised mid-job, AI drafts the variation quote based on the original quote context + new scope description.

⚠️ **Developer Note — AI is always advisory, never autonomous**

The owner must always review and approve AI-generated quotes before they are sent to clients. Build a clear "AI Generated — Review Required" banner on all AI quotes. One-click "approve and send" after review. Log which quotes were AI-generated for analytics.

09 — PRESENTATION

Quote Presentation

CLIENT-FACING QUOTE — REQUIRED ELEMENTS

- OSV Construct branding — Logo, business name, ABN, contact details, licence number
- Quote number & date — e.g. OSV-2026-0042 | Issued: 20 March 2026
- Valid until date — Default 30 days. Configurable per quote.
- Client details — Name, address, job site address if different
- Scope of works — AI-generated plain English summary at the top
- Line items table — Item name, description, qty, unit, price. Materials and labour separated.
- Subtotal, GST, Total inc GST — Clearly formatted
- Exclusions section — What is NOT included. Critical for dispute prevention.
- Terms & conditions — Standard trade terms. Configurable.
- Accept / Decline buttons — Digital approval. Client signs with typed name or e-signature.
- Photos attached — Optional site reference photos included in quote

DELIVERY METHODS

🔗 **Shareable Link**

Unique URL per quote. Client opens in browser — no login required. Mobile-optimised. Accept/decline tracked in real time.

📧 **Email with PDF**

Branded email with PDF attached + link to online version. Auto-populated from quote data. Sent via Resend or SendGrid.

💬 **SMS Link**

"Hi [name], your quote from OSV Construct is ready: [link]" — Sent via Twilio. One tap for client to view and approve.

📄 **PDF Download**

Owner can download branded PDF at any time for printing or manual delivery on site.

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10 — APPROVAL Quote Approval Flow

WHAT HAPPENS WHEN CLIENT ACCEPTS

- Quote status → Approved. Timestamp + client name recorded as acceptance.
- Owner receives instant push notification: *"[Client name] accepted quote OSV-2026-0042 — $8,400"*
- System automatically creates a Customer Ticket linked to the quote
- System prompts owner to schedule the job and assign a tradesperson
- Client receives confirmation email/SMS with expected start date (once scheduled)
- Quote is locked — no further edits. Any changes go through a Variation process.

WHAT HAPPENS WHEN CLIENT DECLINES

- Quote status → Declined. Optional reason field shown to client.
- Lead status → Lost. Decline reason logged for analytics.
- Owner notified. System prompts: "Follow up? Request feedback? Archive?"

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11 — TICKETS Customer Ticket System

A ticket is the operational record of a job. It is created automatically when a quote is approved, or manually by the owner. It is the central hub for everything related to a specific job.

TICKET RECORD — FULL SCHEMA

```
{
  ticket_id:        "UUID",
  ticket_number:    "OSV-T-2026-0087",
  quote_id:         "UUID (linked quote)",
  client_id:        "UUID",
  job_type:         "Decking | Fencing | Carpentry | Landscaping | etc.
  job_address:      "Full site address",
  status:           "Scheduled | In Progress | On Hold | Completed | In
  priority:         "Low | Normal | High | Urgent",
  assigned_to:      "User ID[] (one or more tradies)",
  scheduled_start: "DateTime",
  scheduled_end:   "DateTime",
  actual_start:     "DateTime (set when tradie marks started)",
  actual_end:       "DateTime (set when tradie marks complete)",
  notes:            "Internal job notes",
  photos:           "File[] — before, during, after",
  variations:      "Variation[] (change orders)",
  invoice_id:      "UUID (linked once invoiced)",
  created_at:      "Timestamp"
}
```

TICKET STATUS FLOW

📅 Scheduled → 🔨 In Progress → ⏸️ On Hold → ✅ Completed →

WHAT EACH ROLE SEES ON A TICKET

FIELD / SECTION	OWNER	TRADIE	CLIENT
Job address & details	✅	✅	✅
Assigned tradie	✅	✅	✅ (first name only)
Scheduled dates	✅	✅	✅
Quote / price detail	✅	❌	✅ (own quote)
Internal notes	✅	✅	❌
Job photos	✅	✅	✅ (completion photos)
Variations	✅	Read only	✅ (approve/decline)
Invoice	✅	❌	✅

12 — JOBS Job Workflow & Field Operations

TRADIE MOBILE EXPERIENCE — MUST HAVES

- Today's jobs — Tradie opens app and sees only their assigned jobs for today. Job address with maps link.
- Start job — One tap to mark job started. Timestamps recorded.
- Photo upload — Before, during, and after photos uploaded from phone. Tagged to ticket.
- Notes — Voice-to-text or typed notes added to ticket from site.
- Mark complete — One tap to mark job done. Triggers owner notification.
- Create quote on site — Tradie can start a quote from an existing lead using mobile-optimised quote builder.
- Offline mode — Core functions (view jobs, add notes, upload photos when signal returns) must work without internet.

VARIATION / CHANGE ORDER PROCESS

```
Variation Workflow
```

- Tradie or owner identifies a scope change on site
- Opens ticket → "Add Variation"
- Describes the additional work, adds line items (AI can assist with pricing)
- Variation sent to client for approval before work proceeds
- Client approves/declines via same portal link experience as original quote
- If approved: variation added to job total. Invoice updated automatically.
- If declined: noted on ticket. No additional charge.

13 — COMMS

Communications System

AUTOMATED NOTIFICATIONS — FULL LIST

EVENT	WHO NOTIFIED	CHANNEL	MESSAGE TYPE
New lead received	Owner	Push + Email	Alert
Quote sent to client	Client	Email + SMS	Quote delivery
Quote approved by client	Owner	Push + Email	Win notification
Quote declined by client	Owner	Push	Loss notification
Job scheduled	Client + Tradie	Email + SMS	Confirmation
Job starting tomorrow	Client + Tradie	SMS	Reminder
Job marked complete	Owner + Client	Push + Email	Completion
Invoice sent	Client	Email + SMS	Invoice delivery
Invoice overdue	Client + Owner	Email	Reminder
Variation raised	Client + Owner	Email + SMS	Approval request

14 — DATABASE Database Schema Overview

```
// PostgreSQL — Core tables

users           id, name, email, phone, role, avatar, created_at
clients         id, name, email, phone, address, notes, created_at
leads           id, client_id, source, job_type, suburb, description,
                status, assigned_to, qualification_data, created_at
quotes          id, lead_id, ticket_id, client_id, quote_number, status
                valid_until, subtotal, gst, total, notes,
                internal_notes, ai_generated, created_by, created_at
quote_items     id, quote_id, category, name, description, qty, unit,
                unit_price, cost_price, total, taxable
price_book      id, category, name, description_template, unit,
                default_price, cost_price, margin_pct, active
tickets         id, quote_id, client_id, ticket_number, job_type,
                job_address, status, priority, assigned_to[],
                scheduled_start, scheduled_end, actual_start,
                actual_end, notes, invoice_id, created_at
ticket_photos   id, ticket_id, url, stage (before|during|after), upload
variations      id, ticket_id, description, line_items, status,
                total, client_approved_at, created_by, created_at
invoices        id, ticket_id, client_id, invoice_number, status,
                line_items, subtotal, gst, total, due_date,
                paid_at, created_at
notifications   id, user_id, type, message, read, created_at
follow_ups      id, lead_id, due_at, type, status, draft_message, sent_
```

15 — API API Endpoints

LEADS

POST	/api/leads	Create new lead
GET	/api/leads	List all leads with filters
GET	/api/leads/:id	Get single lead
PUT	/api/leads/:id/status	Update lead status

QUOTES

POST	/api/quotes	Create quote
POST	/api/quotes/ai-generate	Generate quote via AI
GET	/api/quotes/:id	Get quote detail
PUT	/api/quotes/:id	Edit quote (draft only)
POST	/api/quotes/:id/send	Send quote to client
POST	/api/quotes/:id/approve	Client approves quote (public endpoint)
POST	/api/quotes/:id/decline	Client declines quote (public endpoint)
GET	/api/quotes/:id/pdf	Generate & return PDF

TICKETS

POST	/api/tickets	Create ticket
GET	/api/tickets	List tickets with filters

PUT	/api/tickets/:id/status	Update status
POST	/api/tickets/:id/photos	Upload photos
POST	/api/tickets/:id/variations	Add variation

PRICE BOOK & AI

GET	/api/pricebook	Get all price book items
POST	/api/pricebook	Add price book item
PUT	/api/pricebook/:id	Update price
POST	/api/ai/quote	AI quote generation
POST	/api/ai/followup	AI follow-up message draft

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16 — INTEGRATIONS Required Integrations

SERVICE	PURPOSE	PRIORITY
OpenAI GPT-4o	AI quote generation, scope writing, follow-up drafting	P1 AI
Twilio	SMS notifications — quote delivery, job reminders, follow-ups	P1
Resend / SendGrid	Transactional emails — quotes, invoices, confirmations	P1

Stripe	Online invoice payment. Client pays direct from invoice link.	P2
Xero / MYOB	Accounting sync. Push invoices to Xero automatically.	P2
Google Calendar	Job scheduling sync. Owner and tradie calendars.	P2
Cloudflare R2 / S3	Photo and file storage for tickets and quotes	P1 BE
Google Maps API	Address autocomplete and map links in tickets	P2

17 — BUILD ORDER Recommended Build Order

🚀 **Build in this order — each phase delivers standalone value**

Don't build everything at once. Each phase below is usable on its own, generates real business value, and feeds into the next phase.

PHASE 1 — CORE MVP

- Auth system — Owner login. Tradie login. Client link access (no account required).
- Client & lead management — Create, view, update leads. Pipeline kanban view.
- Price book — CRUD interface for owner to manage items.
- Quote builder — Manual quote creation with line items. PDF generation. Send via

email link.

- Quote approval — Client clicks link, views quote, accepts or declines. Owner notified.
- Basic ticket creation — Auto-created on quote approval. Status tracking.

PHASE 2 — AI + OPERATIONS

- AI quote generation — GPT-4o integration. Owner reviews before sending.
- AI follow-up drafting — Automated prompts with pre-written messages for owner to approve.
- Tradie mobile view — Today's jobs. Start/complete job. Photo upload. Notes.
- SMS notifications — Twilio integration. Quote delivery, job reminders.
- Variations system — Change orders mid-job. Client approval flow.
- Invoice generation — Create and send invoices from completed tickets.

PHASE 3 — INTELLIGENCE & GROWTH

- Analytics dashboard — Lead source ROI, quote win rate, revenue by job type, average job value.
- Client portal — Clients log in to see all their jobs, quotes, and invoices in one place.
- Stripe payments — Clients pay invoices online via portal.
- Xero sync — Auto-push invoices to accounting software.
- AI pricing refinement — System learns from won/lost quotes to improve future pricing.
- Review automation — Auto-send Google review request 7 days after job completion.

Critical Problems to Avoid in Build

(1) Never auto-send quotes or messages without owner review. (2) Never show cost/margin data to tradies or clients. (3) Always lock quotes once sent — edits require a new version. (4) Build mobile-first — 80% of tradie usage will be on phone. (5) Every action must have a confirmation — no accidental sends or deletes.

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